

Release Notes for

AT32F402_405 Firmware Library Drivers

V2.0.9-2026/06/02

1. TMR
 - Removed the `tmr_output_channel_fast_set()` and the `tmr_trgout2_enable()`
2. PWC
 - `__force_stores()` is replaced with the `__ISB()` and `__DSB()` to adapt to different compilers
3. CRM
 - Updated the processes of the `crm_reset()`, `crm_hick_sclk_frequency_select()` and `crm_hick_divider_select()` to avoid potential clock failure during switching
4. EXINT
 - Added a new `EXINT_TRIGGER_NONE_EDGE` to adapt to software trigger without edge configuration requirements
5. Updated some notes

V2.0.8-2025/11/24

6. TMR
 - Updated these functions: `tmr_pwm_input_config()`, `tmr_encoder_mode_config()` and `tmr_output_channel_config()`
The channel must be disabled before configuring channel mode, otherwise it fails to switch between channel modes
7. Updated some notes

V2.0.7-2025/06/25

8. CRM
 - Modified bit field “`adc14rst`” to “`tmr14rst`”.
9. CAN
 - Modified definitions of some register bit fields, with 14 filter banks only, and deleted redundant definitions.
10. Updated some notes.

V2.0.6-2025/01/20

1. PWC
 - Updated LDO output level settings, removed LDO configuration option “`PWC_LDO_OUTPUT_1V0`” and added “`PWC_LDO_OUTPUT_1V1`”
2. TMR
 - Updated “`tmr_brk_filter_value_set()`” function, and modified bit field “`bkf`” to “`brkf`”.
3. Updated some notes.

V2.0.5-2024/12/13

1. CAN

- Updated “can_transmit_cancel()” function writing to avoid cross-mailbox interference due to misoperation.
2. USB
 - Updated “usb_hch_halt()” function in host mode and improved channel halt processing flow.
 - Updated “usb_hc_enable()” function in host mode to ensure that its original state is cleared when enabling the corresponding channel.
 - Updated “usb_host_disable()” function in host mode, and updated process of disabling channels to avoid inadvertently opening the remaining channels.
 3. SPI
 - Updated frequency calculation method in the “i2s_init()” function, and fixed the error caused by calculation value overflow.
 4. ERTC
 - Updated “ertc_wakeup_enable()” function, and corrected the use of parameters.
 - Updated “ertc_wait_update()” function, and standardized how this function is written to work across all series.
 5. Updated some notes.

V2.0.4-2024/08/08

1. PWC
 - Updated “pwc_voltage_regulate_set()” function, and removed “PWC_REGULATOR_LOW_POWER” setting.
2. QSPI
 - Added “qspi_auto_ispc_enable()” function for bus input clock phase matching.
 - Revised FIFO depth definition from “QSPI_DMA_FIFO_THOD_WORD32” to “QSPI_DMA_FIFO_THOD_WORD24”.
 - Updated “qspi_flag_clear()” function writing, and fixed AC6 compilation problems.
3. SCFG
 - Adjusted the type of return value of “scfg_sram_operr_status_get()” function from “error_status” to “flag_status”.
4. USART
 - Updated “usart_init()” function and revised UART7/UART8 clock source.
5. USB
 - Updated “usb_hc_enable()” function in host mode, and fixed the enumeration issue of some devices.
 - Updated “usb_host_disable()” function in host mode, and fixed the issue that channels might not close properly in DMA mode.
 - Revised parameters in “usb_read_packet()” function, and fixed AC6 compilation problems.
6. Updated some notes.

V2.0.3-2024/01/05

1. Updated USB driver timeout handling.
2. Adjusted USART DE line polarity “usart_de_polarity_set()” function.
3. Removed the configuration of serial signal as infrared modulation signal source.
4. Updated some notes.

V2.0.2-2023/10/26

1. Added “interrupt_flag_get()” function to each of the IP sections.

V2.0.1-2023/09/28

1. Updated USB driver.

V2.0.0-2023/08/18

1. Initial release.